

I understand the correction. This plan will focus only on the uploaded close field agricultural vision foundation model, its current DINOv3 style and MIM codebase, the roughly 1.87 million image corpus, and the evidence needed for a credible CVPR submission.

Thought for 1m 14s >

CVPR Research Plan for the Close Field Agricultural Vision Foundation Model

This plan applies only to your uploaded agricultural foundation model project. It does not concern the PlantNaturalist mobile application.

Your current foundation is already useful.

- Approximately 1.87 million close field agricultural images.
- More than 70 datasets or folders.
- Strong coverage of plant species, diseases, weeds, seedlings, and crops.
- Official ViT S, ViT B, and ViT L backbones.
- ImageNet, MAE, DINOv2, and DINOv3 initialization.
- MIM and teacher student self supervised training.
- Reproducible configurations, checkpoints, logs, and manifests.

 architecture

 dataset-strategy

The missing part is not simply more pretraining. The project needs one defensible method contribution, contamination free benchmarks, realistic domain shifts, and consistent downstream evaluation.

No experiment plan can guarantee CVPR acceptance. The objective is to remove the obvious reasons reviewers could reject the work.

1. Fixed scientific identity

Recommended title

CloseAgriFM, Morphology Aware Continual Pretraining for Close Field Agricultural Vision

Primary scope

The paper should cover close field agricultural imagery.

Included:

- Individual plants.
- Leaves, flowers, fruits, stems, and seeds.
- Diseases and nutrient stress.
- Crops and weeds.
- Seedlings and growth stages.
- Handheld cameras.
- Field robots.
- Tractor mounted cameras.
- Greenhouses.
- Orchards.
- Ground level field imagery.

Do not make satellite or large scale remote sensing central to this paper. Your current 1.87 million image corpus mainly contains close field RGB data. Mixing satellite, UAV mapping, and close plant imagery would weaken the scientific scope rather than broaden it.

 dataset-strategy

Primary claim

CloseAgriFM learns agricultural representations that transfer across species, diseases, crops, fields, tasks, and label budgets better than general vision foundation models and existing agricultural pretraining.

Secondary claim

Morphology aware masked learning complements DINOv3 style self distillation by preserving local structures such as leaf boundaries, lesions, stems, flowers, fruits, and small weeds.

Optional sensor claim

Keep the existing one by one spectral adapter as an engineering feature. Do not make sensor adaptation a major contribution unless you add substantial real close range multispectral data from multiple sensors.

Synthetic five band examples cannot support a multispectral scientific claim. Your documentation correctly identifies synthetic demo data as engineering verification rather than research evidence.

 data-formats

2. Final model selection

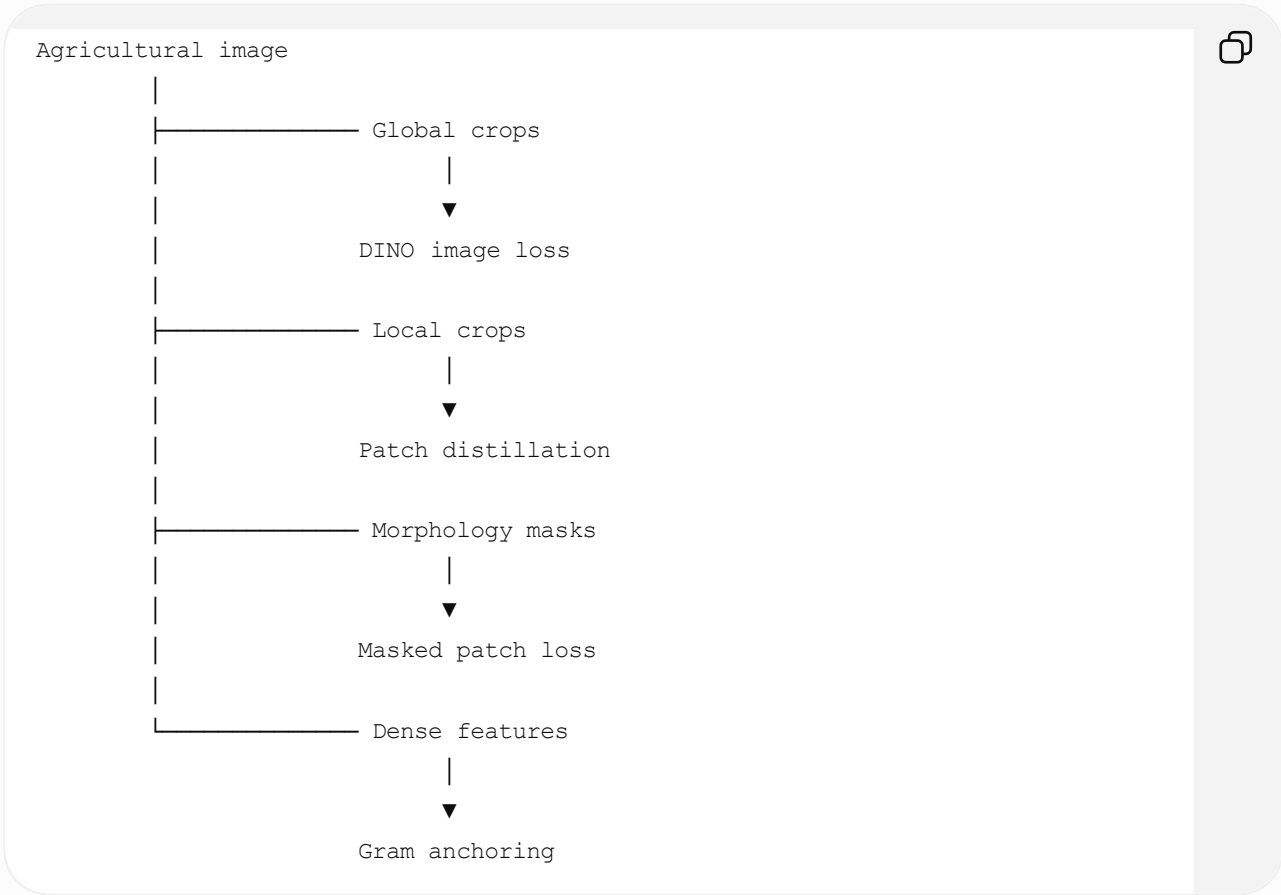
Main model

DINOv3 ViT B 16 initialized CloseAgriFM

Use ViT S for method development and ablations. Use ViT B for the principal paper model. Train ViT L only for one scaling experiment if compute permits.

DINOv3 is the appropriate starting point because it targets both global representation quality and dense spatial features. It introduced Gram anchoring to reduce the degradation of dense patch representations during long self supervised training. [arXiv +1](#)

Final pretraining objective



Use the following loss.

$$L = L_{\text{DINO}} + L_{\text{iBOT}} + \lambda_m L_{\text{morphology}} + \lambda_g L_{\text{Gram}}$$

Recommended initial weights:

Loss	Weight
DINO global	1.0
iBOT patch	1.0
Morphology MIM	0.5
Gram anchor	0.1

These values begin the ViT S experiments. The validation study should determine the final weights.

3. Method contribution, morphology aware masking

Generic random masking may remove background, plant tissue, and biologically useful structures without distinction.

The proposed agricultural component should create three mask families.

Mask A, random block

Standard block masks provide the control condition.

Mask B, foreground concentrated

Estimate plant foreground without labels using teacher attention or unsupervised saliency.

Sample more masked patches from likely plant regions, while retaining some background masks.

Mask C, boundary and detail concentrated

Use image gradients and multiscale edge responses to emphasize:

- Leaf boundaries.
- Thin stems.
- Lesion edges.
- Flower structures.
- Fruit boundaries.
- Weed crop intersections.
- Venation and texture regions.

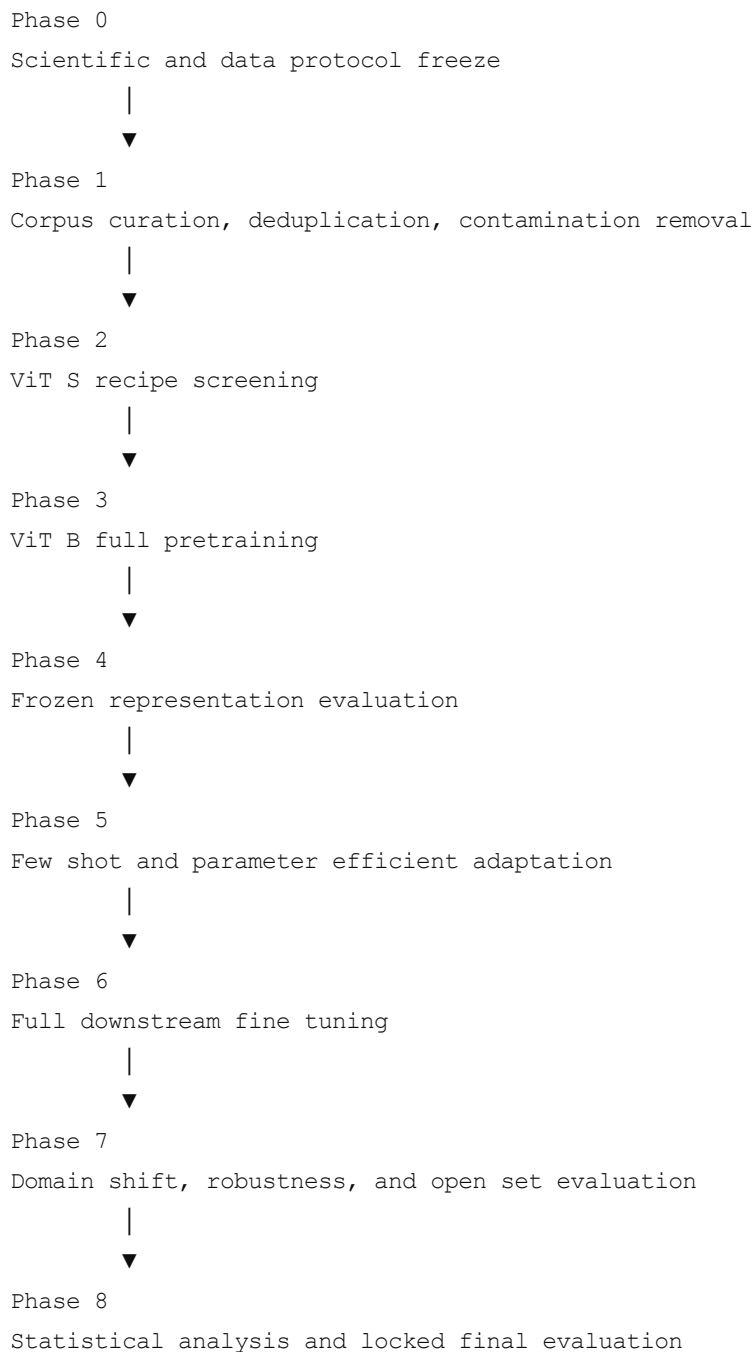
Mixed mask schedule

Use a stochastic mixture.

Mask family	Probability
Random block	0.40
Foreground	0.35
Boundary detail	0.25

This method needs ablation against standard random masking. Do not assume it helps. It becomes a contribution only when it improves both global and dense downstream tasks.

4. Project level experiment map



5. Phase 0, freeze the research protocol

Complete this before expensive training.

5.1 Define the model variants

Use immutable identifiers.

ID	Model
G0	DINOv3 ViT B, unchanged
G1	Agricultural MIM only
G2	Agricultural DINO only
G3	DINO plus iBOT
G4	DINO plus iBOT plus morphology masking
G5	G4 plus Gram anchoring
G6	Sequential MIM then DINO
Final	Best validated variant

G5 is the proposed main method. G0 to G4 and G6 are scientific controls.

5.2 Predefine the primary outcomes

Do not choose metrics after seeing results.

Task	Primary metric
Species classification	Macro F1
Disease classification	Macro F1
Object detection	COCO AP
Semantic segmentation	mIoU
Instance segmentation	Mask AP
Counting	MAE
Retrieval	mAP at 10
Open set recognition	AUROC
Domain shift	Worst domain score

5.3 Lock the evaluation rules

- No downstream validation or test images in pretraining.
- No near duplicates across pretraining and downstream test sets.
- No test set tuning.
- Same task head for every backbone.
- Same augmentation and optimization budget.
- Same label subsets across models.
- Three downstream seeds.
- Five label samplings for low label evaluation.
- Report every failed or unstable experiment.

6. Phase 1, corpus curation

The current corpus has strong RGB volume, but its quality and diversity matter more than the raw count. The uploaded audit identifies duplicate analysis, licensing, frozen source versions, and external transfer datasets as unfinished work.

 dataset-strategy

6.1 Build corpus versions

Create several frozen versions rather than one changing folder.

Version	Description
C0	Raw corpus
C1	Corrupt files removed
C2	Exact duplicates removed
C3	Near duplicates removed
C4	Benchmark contamination removed
C5	Source balanced corpus
C6	Final publication corpus

Every reported model should reference a corpus manifest hash.

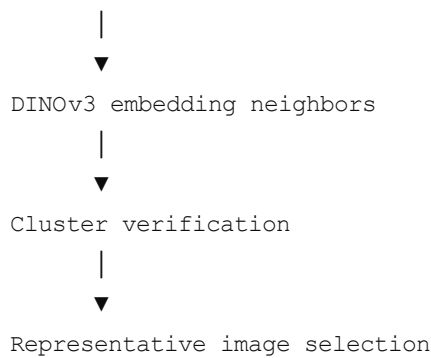
6.2 Deduplication pipeline

Exact SHA 256



Perceptual hash





Use source aware rules. Repeated images across Kaggle copies should remain one duplicate group even when filenames differ.

6.3 Benchmark contamination audit

Compare the complete pretraining corpus against every downstream validation and test set.

Match by:

- Exact checksum.
- Perceptual hash.
- Filename and source identifier.
- Image dimensions.
- Embedding similarity.
- Cropped duplicate similarity.
- Observation or sequence identifier.

PlantNet 300K currently contributes roughly 306,000 images to your corpus. If PlantNet becomes a strict downstream benchmark, remove the full dataset from pretraining.

Otherwise, call its evaluation transductive rather than independent. [dataset-strategy](#)

6.4 Source balancing

PlantNet and a few disease datasets could dominate the representation.

Sample source s using:

$$P(s) \propto n_s^\alpha$$

Evaluate:

Sampling	Alpha
Proportional	1.0
Square root	0.5

Sampling	Alpha
Uniform source	0.0

Use square root balancing as the starting choice.

6.5 Content balance

Target broad representation across:

Category	Target range
Species and taxonomy	20 to 25 percent
Disease and stress	20 to 25 percent
Crop and weed	15 to 20 percent
Field canopy	10 to 15 percent
Seedling and growth	10 percent
Fruit and flower	10 percent
Other close field	5 to 10 percent

Do not duplicate images merely to reach these percentages. Control effective sampling weights instead.

6.6 Reserve external datasets

Hold out entire datasets for downstream evaluation.

Recommended strict holdouts:

- PlantNet 300K.
- PlantDoc.
- Global Wheat Head Dataset 2021.
- PhenoBench.
- MinneApple.
- A newly collected external field set.

These datasets must not contribute unlabeled images to the strict pretraining corpus.

7. Phase 2, ViT S recipe screening

Do not begin with a full ViT B run.

Use ViT S and approximately 25 percent of C6 to determine the method.

7.1 Screening configuration

Setting	Value
Backbone	DINOv3 ViT S 16
Data	25 percent of C6
Resolution	224
Global crops	2
Local crops	6
Global batch	1,024
Precision	BF16
Steps	75,000
Warmup	7,500
Optimizer	AdamW
Seeds	3

7.2 Objective experiments

Run:

Run	Global	Patch	Morphology	Gram
P0	DINOv3 baseline	Yes	No	Existing
P1	No	MIM	No	No
P2	DINO	No	No	No
P3	DINO	iBOT	No	No
P4	DINO	iBOT	Yes	No
P5	DINO	iBOT	Yes	Yes

Run	Global	Patch	Morphology	Gram
P6	MIM then DINO	iBOT	No	Yes

P5 should not automatically win. Select using held out sentinel tasks.

7.3 Masking experiments

Test:

- Mask ratios, 0.4, 0.6, and 0.75.
- Random masks.
- Foreground masks.
- Boundary masks.
- Mixed morphology masks.
- Fixed mask distribution.
- Curriculum masking.

Curriculum option:

```

Early training
Mostly random masks
    |
    ▼
Middle training
Random plus foreground
    |
    ▼
Late training
Random plus foreground plus detail masks

```



7.4 Crop experiments

Test:

Setting	Values
Global crops	2
Local crops	4, 6, 8
Global scale	0.4 to 1.0
Local scale	0.05 to 0.4
Resolution	224, 320

Setting	Values
Teacher momentum	Constant, cosine

Your project documentation currently describes DINO style training, while another roadmap describes DINOv3 style features such as Gram anchoring. Before the publication run, verify the actual implementation and use one precise method name throughout the code and paper. [research-roadmap 2](#) [architecture](#)

8. Sentinel validation during pretraining

SSL loss alone should never select the foundation checkpoint.

Four sentinel tasks

Reserve small validation sets for:

1. Fine grained species classification.
2. In the wild disease classification.
3. Crop weed semantic segmentation.
4. Plant organ detection.

Run frozen evaluation every 10,000 steps.

Checkpoint score

Normalize each metric relative to the unchanged DINOv3 baseline.

$$S = 0.25S_{\text{species}} + 0.25S_{\text{disease}} + 0.25S_{\text{segmentation}} + 0.25S_{\text{detection}}$$

Select checkpoints using this validation score.

Representation diagnostics

Track:

- Feature standard deviation.
- Effective rank.
- Singular value distribution.
- Teacher entropy.
- Student entropy.
- Patch variance.
- CLS token norm.
- Patch token norm.
- Nearest neighbor purity.

- Foreground background separation.
- Dense correspondence accuracy.
- Gram similarity.
- Source predictability.
- Dataset predictability.

High dataset predictability may indicate that the model learned source watermarks, image sizes, or backgrounds instead of biological structure.

9. Phase 3, full ViT B pretraining

After ViT S screening, train only the best recipe and one strong control.

Main comparison

- DINOv3 ViT B unchanged.
- DINOv3 ViT B with ordinary agricultural DINO plus iBOT.
- CloseAgriFM ViT B with morphology masking and Gram anchoring.

Recommended starting schedule

Setting	Value
Backbone	DINOv3 ViT B 16
Corpus	C6
Main resolution	224
Global crops	2
Local crops	8
Global batch	2,048
Main steps	250,000
Warmup	20,000
Precision	BF16
Optimizer	AdamW
Weight decay	0.04 to 0.4

Setting	Value
Gradient clipping	3.0
Checkpoint interval	5,000 steps
After main pretraining:	
Stage	Steps
Gram anchoring	20,000
High resolution adaptation	20,000
High resolution	448

DINOv3 used separate pretraining, Gram anchoring, and high resolution adaptation stages. Your work should describe its implementation as domain continual pretraining, not as a reproduction of the original DINOv3 scale. [arXiv +1](#)

Repetition

Preferred:

- Two independent full pretraining seeds.
- Three downstream seeds per checkpoint.

Minimum:

- One full ViT B pretraining.
- Three ViT S pretraining seeds.
- Three downstream seeds for every main comparison.

10. Benchmark suite

Call the evaluation suite **CloseAgriBench**.

It should cover global recognition, dense prediction, localization, counting, retrieval, label efficiency, and domain shift.

CloseAgriBench

```

|
|— Classification
|— Detection
|— Semantic segmentation
|— Instance segmentation

```



11. Core downstream datasets

11.1 PlantNet 300K

Purpose:

- Fine grained species recognition.
- Long tailed classification.
- Retrieval.
- Rare class transfer.
- Taxonomic confusion.

Strict condition:

Remove PlantNet completely from the pretraining corpus.

Evaluation:

- Top 1.
- Top 5.
- Macro F1.
- Mean class recall.
- Balanced accuracy.
- Retrieval mAP.

11.2 PlantDoc

Purpose:

- In the wild disease recognition.
- Natural backgrounds.
- Cross dataset disease transfer.
- Open set disease detection.

PlantDoc contains 2,598 images across 13 plant species and multiple healthy or disease classes. It was designed around nonlaboratory plant disease imagery. 

Evaluation:

- Macro F1.
- Balanced accuracy.

- Per disease recall.
- ECE.
- Open set AUROC.

11.3 Global Wheat Head Dataset 2021

Purpose:

- Small object detection.
- Counting.
- Country and acquisition shift.
- Field level generalization.

GWHD 2021 expanded the earlier dataset with additional countries, images, and annotations. The benchmark supports realistic cross-domain wheat head localization.



Evaluation:

- COCO AP.
- AP50.
- AP75.
- AP small.
- Count MAE.
- Country held out AP.
- Worst acquisition session AP.

11.4 PhenoBench

Purpose:

- Crop weed semantic segmentation.
- Plant instance segmentation.
- Leaf instance segmentation.
- Unseen field evaluation.

PhenoBench provides pixel labels for crops and weeds, plant and leaf instances, and a hidden test set containing an unseen field.  arXiv +1

Evaluation:

- mIoU.
- Crop IoU.
- Weed IoU.
- Boundary F1.
- Panoptic quality.
- Plant instance AP.

- Leaf instance AP.

11.5 MinneApple

Purpose:

- Fruit detection.
- Instance segmentation.
- Counting.
- Orchard background transfer.
- Occlusion analysis.

MinneApple contains 1,000 orchard images and more than 41,000 apple instances with polygon masks.  [arXiv](#)

Evaluation:

- Box AP.
- Mask AP.
- AP small.
- Count MAE.
- Count RMSE.
- Occlusion stratified AP.

11.6 External close field benchmark

Public benchmarks alone are not enough for a strong domain generalization claim.

Collect a small but carefully controlled external benchmark.

Minimum design:

Dimension	Minimum
Farms or sites	4
Crops	5
Cameras	3
Seasons or dates	2
RGB images	10,000
Dense annotations	1,000
Detection annotations	2,000

Dimension	Minimum
Classification labels	5,000

The external set must remain absent from pretraining and method development.

12. Downstream training protocols

Each task should have four adaptation tracks.

Track A, frozen linear probing

Purpose:

Measure representation quality without changing the encoder.

Train:

- One linear layer.
- Frozen backbone.
- Frozen adapter.
- No heavy augmentation.
- Three seeds.

Classification pooling comparison:

- CLS token.
- Mean patch token.
- CLS plus mean patch token.

Select pooling on validation only.

Track B, frozen dense probing

Purpose:

Measure spatial representation quality.

Use:

- Frozen encoder.
- One by one segmentation head.
- Final patch features.
- Bilinear upsampling.

A second protocol can aggregate the last four transformer blocks using fixed concatenation.

Track C, parameter efficient tuning

Use LoRA.

Evaluate ranks:

- 4.
- 8.
- 16.

Recommended primary result:

- LoRA rank 8.
- Trainable task head.
- Trainable normalization.
- Less than 5 percent trainable parameters.

Track D, full fine tuning

Train the complete backbone and task head.

This measures application performance, but it should not replace frozen evaluation.

13. Label efficiency protocol

Every core task should use several annotation budgets.

Budget	Fraction
Very low	1 percent
Low	5 percent
Moderate	10 percent
Partial	25 percent
Medium	50 percent
Full	100 percent

For small datasets, also use examples per class:

- 1 shot.

- 5 shot.
- 10 shot.
- 25 shot.

Use five independently sampled subsets at each low label setting.

Report:

- Mean.
- Standard deviation.
- 95 percent confidence interval.
- Area under the label efficiency curve.
- Labels needed to match DINOv3 full data performance.
- Performance retention at 10 percent labels.

Agricultural foundation models already evaluate broad downstream transfer and label efficiency. Agri FM Plus used close field agricultural continual pretraining, FoMo4Wheat evaluated ten in-field tasks using 2.5 million images from 30 sites, and SPROUT reported a 2.6 million image multi-crop foundation model. A CVPR submission must go beyond a single downstream task or a simple ImageNet comparison. [GyF CVF Open Acc... +2](#)

14. Standard downstream heads

Use the same heads for every backbone.

Classification

- Linear head for probing.
- Standard linear head for full fine tuning.

Detection

Use ViTDet with Faster R CNN.

Alternative:

Use a simple feature pyramid with Cascade R CNN, but apply it to every ViT backbone.

Semantic segmentation

- Linear decoder for frozen probing.
- UPerNet for full fine tuning.

Instance and panoptic segmentation

Use Mask2Former.

Counting

Evaluate:

- Count from detected instances.
- Direct regression from CLS token.
- Density map prediction from patch features.

Retrieval

Use normalized embeddings and cosine similarity.

No task-specific training in the primary retrieval protocol.

15. Fair downstream optimization

Classification schedule

Setting	Linear	Full
Epochs	50	100
Optimizer	AdamW	AdamW
LR trials	3	3
Seeds	3	3

Suggested learning rates:

- Linear, 0.001, 0.003, 0.01.
- Full, 0.000001, 0.00001, 0.00003.

Detection schedule

- 24 epoch standard schedule.
- 36 epoch secondary schedule.
- Same image resolution.
- Same augmentations.
- Same feature pyramid.
- Same learning rate search.

Segmentation schedule

- 80,000 iterations.
- Same crop size.
- Same decoder.

- Same batch size where possible.
- Poly learning rate decay.
- Validation every 4,000 iterations.

Hyperparameter policy

Every baseline receives:

- The same number of trials.
- The same training budget.
- The same augmentation policy.
- The same early stopping rules.
- The same task head.
- The same test resolution.

16. Baseline model families

Capacity matched primary baselines

Baseline	Scale
ImageNet ViT B	Matched
MAE ViT B	Matched
DINOv2 ViT B	Matched
DINOv3 ViT B	Matched
CloseAgriFM ViT B	Matched

Agricultural baselines

Include when public checkpoints and licenses permit:

- Agri FM Plus.
- FoMo4Wheat.
- SPROUT.

Agri FM Plus appeared at the CVPR 2025 Vision for Agriculture workshop rather than the CVPR main conference. It remains a necessary close field comparison because it uses a two-stage agricultural continual learning approach. [CyF CVF Open Acc... +1](#)

Broader representation baselines

Use on representative tasks:

- CLIP.
- SigLIP 2.
- ConvNeXt ImageNet.
- ResNet 50 ImageNet.
- SAM encoder where compatible.

Do not force all architectures into a ViT-specific head when that would be technically inappropriate. Report those comparisons separately.

17. Domain shift experiments

17.1 Source held out

Train on all but one dataset source.

Test on the excluded source.

17.2 Crop held out

Hold out complete crop families.

Example:

```
Train
Rice, maize, cotton, tomato, soybean

Test
Wheat
```



Rotate the held out family.

17.3 Geography held out

For datasets with location metadata:

- Hold out country.
- Hold out farm.
- Hold out field.
- Hold out acquisition session.

GWHD is useful because it contains diverse countries and acquisition conditions.



17.4 Time held out

Train on one season or date.

Test on later dates.

17.5 Device held out

Train on:

- DSLR.
- Web images.
- One smartphone family.

Test on:

- Unseen smartphone.
- Field robot.
- Different focal length.
- Different compression pipeline.

17.6 Background held out

Evaluate separately on:

- Laboratory background.
- Soil.
- Dense canopy.
- Greenhouse.
- Orchard.
- Natural mixed vegetation.

Domain metrics

Report:

$$\text{Retention} = \frac{\text{OOD score}}{\text{In-domain score}}$$

Also report:

- Mean domain performance.
- Worst domain performance.
- Standard deviation across domains.
- Absolute OOD drop.
- Relative OOD drop.

The worst domain score should be a primary result.

18. Cross-dataset transfer experiments

These experiments directly test representation transfer.

Disease transfer

Train on controlled disease images



Test on PlantDoc natural images



Weed segmentation transfer

Train on PhenoBench fields



Test on an external field dataset



Detection transfer

Train on selected GWHD countries



Test on held-out countries



Orchard transfer

Train on MinneApple



Test on a separately collected orchard



Evaluate zero-shot transfer, five-shot adaptation, and full adaptation.

19. Retrieval and clustering evaluation

Foundation models should support more than supervised heads.

Retrieval tasks

- Same species.
- Same genus.
- Same disease.
- Same crop.
- Same plant organ.
- Same growth stage.
- Same object across viewpoints.

Metrics

- Recall at 1.
- Recall at 5.
- Recall at 10.
- mAP at 10.
- mAP at 100.
- Normalized mutual information.
- Adjusted Rand index.

Hard retrieval sets

Create query galleries where:

- Backgrounds differ.
- Species look similar.
- Diseases share symptoms.
- Images come from different datasets.
- Camera quality differs.

This helps determine whether the representation learned biology or source appearance.

20. Open set recognition

Real agricultural systems encounter species and diseases absent from training.

Species holdout

Hold out 10 percent of species completely.

Disease holdout

Hold out entire disease categories.

Crop holdout

Hold out one crop family.

Methods

Compare:

- Maximum softmax.
- Energy score.
- k nearest neighbor distance.
- Class prototype distance.
- Mahalanobis distance.

Metrics

- AUROC.
 - AUPR.
 - FPR95.
 - Open set classification rate.
 - Accuracy at fixed coverage.
-

21. Robustness benchmark

Create **CloseAgri C**, an agricultural corruption suite.

Camera corruptions

- Defocus blur.
- Motion blur.
- JPEG compression.
- Sensor noise.
- Underexposure.
- Overexposure.
- White balance shift.
- Lens flare.

Field corruptions

- Dust.
- Fog.
- Rain.
- Water droplets.
- Soil splatter.
- Partial leaf occlusion.
- Wind motion.
- Mixed vegetation.

Biological variation

- Early growth.
- Senescence.
- Disease severity.
- Leaf damage.
- Overlapping leaves.
- Dense fruit clusters.
- Small weeds.

Use five severity levels.

Report:

- Mean corruption error.
 - Relative corruption error.
 - Worst corruption score.
 - Calibration under corruption.
-

22. Calibration and selective prediction

Report confidence quality, not only accuracy.

Classification

- Expected calibration error.
- Adaptive calibration error.
- Negative log likelihood.
- Brier score.
- Reliability diagram.

Detection

- Detection ECE.
- Precision at fixed confidence.
- Recall at fixed false detections per image.

Segmentation

- Pixel calibration error.
- Selective mIoU.
- Risk coverage curve.

Abstention

Evaluate predictions at:

Coverage	Meaning
100 percent	No rejection
90 percent	Reject 10 percent
80 percent	Reject 20 percent
70 percent	Reject 30 percent

A good representation should become substantially more accurate when uncertain cases are rejected.

23. Main ablation program

A. Initialization

- Random.
- ImageNet.
- MAE.
- DINOv2.
- DINOv3.
- DINOv3 plus agricultural continuation.

B. Objective

- MIM only.
- DINO only.
- DINO plus iBOT.
- Sequential MIM then DINO.
- Joint objective.
- Joint objective plus Gram anchoring.

C. Mask design

- Random.
- Foreground.
- Boundary detail.
- Mixed morphology.
- Curriculum morphology.

D. Mask ratio

- 0.40.
- 0.60.
- 0.75.

E. Data scale

- 100,000.
- 250,000.
- 500,000.
- 900,000.
- Full C6.

F. Data diversity

- Species dominant.
- Disease dominant.
- Field dominant.
- Balanced.

G. Source sampling

- Proportional.
- Square root.
- Uniform.

H. Curation

- Raw.
- Exact deduplication.
- Near duplicate removal.
- Full contamination removal.

I. Resolution

- 224.
- 320.
- 448.

J. Backbone scale

- ViT S.
- ViT B.
- ViT L.

K. Adaptation

- Frozen linear.
- LoRA.
- Decoder only.
- Full fine tuning.

24. Statistical protocol

Main experiments

Use three seeds:

- 7.
- 21.
- 42.

Low label experiments

Use five different sampled label subsets.

Use the same subsets for every model.

Confidence intervals

Report 95 percent bootstrap confidence intervals.

Bootstrap using the correct unit:

- Image for independent images.
- Plant for repeated plant images.
- Field for field datasets.
- Country for geographic experiments.
- Sequence for video or burst images.

Tests

Use:

- Paired bootstrap for prediction comparisons.
- Wilcoxon signed rank across datasets.
- Holm correction for multiple comparisons.
- Effect sizes with p values.

Do not declare superiority from one favorable seed.

25. Model selection and test locking

Pretraining selection

Use sentinel validation only.

Downstream selection

Use each dataset's validation split.

Test policy

1. Freeze code.
2. Freeze manifests.
3. Freeze model checkpoints.
4. Freeze hyperparameters.
5. Freeze confidence thresholds.
6. Generate a release candidate tag.
7. Run test evaluation.
8. Store predictions without changing the model.

For hidden benchmarks such as PhenoBench, limit submissions during model development. Its unseen-field test design makes it especially useful for generalization evidence.  arXiv +1

26. Engineering work required before experiments

Your current implementation is scientifically promising but still has scaling limitations.

The documentation states that the launchers remain single process, distributed behavior requires an externally initialized process group, and automatic gradient accumulation is absent.  training

Implement:

- `torchrun` distributed launch.
- Distributed checkpoint safety.
- Gradient accumulation.
- Global batch accounting.
- Source balanced sampler.
- Dataset mixture sampler.
- iBOT patch objective.
- Morphology mask generator.
- Verified Gram anchoring.
- High resolution adaptation.
- Downstream model registry.
- Benchmark split registry.
- Automatic contamination reports.
- Aggregate seed analysis.
- Bootstrap confidence intervals.
- Compute and energy accounting.

Your current implementation map has clean boundaries between data, adapters, backbones, objectives, and training. This should permit the new losses and downstream heads without rewriting the loader or ViT implementation.  implementation-map

27. Experiment tracking requirements

Your current project already records commands, configurations, Git information, environments, metrics, checkpoints, and random state.

Add these publication fields:

- Corpus version.
- Corpus manifest hash.
- Deduplication version.
- Downstream split hash.
- Benchmark contamination status.
- Number of images seen.
- Number of tokens seen.
- Effective batch size.
- GPU model.
- GPU count.
- GPU hours.
- Energy estimate.
- Peak memory.
- Throughput.
- Failed run reason.
- Final prediction file hash.

CVPR 2026 introduced mandatory basic compute reporting and encouraged more complete compute transparency. Preserve these records from the beginning rather than reconstructing them near submission.  CVPR +1

28. Primary paper tables

Table 1, corpus

Include:

- Images.
- Sources.
- Species.

- Crops.
- Disease groups.
- Countries.
- Camera types.
- Duplicate removal.
- Excluded images.
- Licenses.

Table 2, frozen representation

Include:

- Species linear probe.
- Disease linear probe.
- Segmentation linear probe.
- Retrieval.
- Counting.

Table 3, full downstream transfer

Include:

- Classification.
- Detection.
- Semantic segmentation.
- Instance segmentation.
- Counting.

Table 4, label efficiency

Include 1, 5, 10, 25, and 100 percent results.

Table 5, domain shift

Include:

- Source held out.
- Crop held out.
- Field held out.
- Geography held out.
- Device held out.

Table 6, ablations

Include:

- Objective.
- Masking.

- Gram anchoring.
- Sampling.
- Deduplication.
- Data scale.

Table 7, compute

Include:

- Parameters.
 - FLOPs.
 - Training images.
 - Tokens.
 - GPU hours.
 - Throughput.
 - Memory.
 - Average benchmark rank.
-

29. Primary figures

Figure 1, method

Show the DINOv3 teacher student pipeline, iBOT loss, morphology masks, and Gram anchor.

Figure 2, corpus

Show dataset balance, crops, disease groups, image sources, and geographic coverage.

Figure 3, mask examples

Show:

- Random masks.
- Foreground masks.
- Boundary masks.
- Reconstructed or predicted regions.

Figure 4, label efficiency

Plot metric against labeled data percentage for classification, detection, and segmentation.

Figure 5, domain shift

Use a heatmap with models as rows and held-out domains as columns.

Figure 6, feature maps

Show PCA patch features for:

- Crop.
- Weed.
- Soil.
- Fruit.
- Leaf.
- Lesion.
- Background.

Figure 7, scaling

Show performance against:

- Dataset size.
- Model size.
- GPU hours.

Figure 8, failures

Show difficult cases and explain the failure source.

30. Go or no go criteria

These are research targets, not guaranteed acceptance thresholds.

Continue toward CVPR when

- CloseAgriFM beats DINOv3 on the average benchmark rank.
- Gains appear in at least four task families.
- Morphology masking improves dense prediction.
- Gram anchoring prevents dense feature degradation.
- Low label gains remain across multiple datasets.
- Out-of-domain gains exceed in-domain-only gains.
- Results reproduce across seeds.
- The strict contamination audit passes.

Reconsider the main conference when

- Gains occur only on disease classification.
- Full fine tuning gains disappear under frozen probing.
- Performance improves only on datasets present in pretraining.
- The method loses to ordinary DINOv3 under unseen fields.
- Morphology masking has no measurable benefit.

- Dataset diversity explains all gains.
- No external field benchmark exists.
- The paper depends on one favorable seed.

In those cases, the work may still fit a CVPR workshop, WACV, Plant Phenomics, or Computers and Electronics in Agriculture, but it would need a more conservative contribution statement.

31. Recommended execution order

Months 1 and 2

- Corpus C0 to C6.
- License audit.
- Deduplication.
- Benchmark contamination removal.
- Strict downstream holdouts.
- Distributed training implementation.

Months 3 and 4

- ViT S objective experiments.
- Mask ratio experiments.
- Morphology masking experiments.
- Gram anchoring verification.
- Sentinel task framework.

Months 5 and 6

- ViT B full pretraining.
- High resolution adaptation.
- Second final pretraining seed where possible.
- Frozen representation evaluation.

Months 7 and 8

- Classification.
- Detection.
- Semantic segmentation.
- Instance segmentation.
- Counting.
- Retrieval.

Months 9 and 10

- Label efficiency.
- Domain shift.
- Open set recognition.
- Corruption robustness.
- Calibration.

Month 11

- Statistical analysis.
- Reproduction runs.
- Compute accounting.
- Final locked test evaluation.

Month 12

- Paper.
- Supplement.
- Anonymous code.
- Model cards.
- Dataset statements.
- Failure analysis.
- Rebuttal preparation.

32. Final contribution package

The final submission should make four claims, no more.

1. Method

Morphology aware DINOv3 continual pretraining improves both global and dense agricultural representations.

2. Data

A provenance controlled, deduplicated, and source balanced corpus contains approximately 1.87 million close field agricultural images.

3. Benchmark

CloseAgriBench measures classification, detection, segmentation, counting, retrieval, label efficiency, and realistic domain shift.

4. Evidence

CloseAgriFM outperforms general and agricultural foundation models under frozen, low label, parameter efficient, full fine tuning, and unseen-domain protocols.

Your software tests already cover loaders, official backbones, adapters, MIM and DINO behavior, checkpoints, resumes, configurations, and deterministic state. Those tests establish engineering reliability, not representation quality.  testing

The CVPR paper will succeed or fail on the scientific evidence added above, especially strict contamination control, frozen dense evaluation, morphology-aware ablations, low-label transfer, and unseen-field performance.